

FriendOrFoe

The three dependencies the study varies, and the measure that is supposed to tell them apart

“The experiment CHART was built for, and the first to run on it.”

website, FriendOrFoe, opening

The question is written on the page in one sentence.

“Whether role dependencies raise complementarity in a human-AI defense team facing zero-day attacks, and which of the leader, pool and informational dependencies carry the effect.”

website, FriendOrFoe, The question

“The design is fixed and the platform exists. Data collection is what remains.”

website, FriendOrFoe, The question

That is the reason this file matters more than a lexicon usually would. Nothing here can be settled later by looking at results, because the results do not exist yet, and every term in the pre-registration will be read against the design after collection rather than before it. Three of the terms below do not mean what the platform means by the same word, and one of them, leader, is a name the platform does not have.

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website, FriendOrFoe, Support

Contents

1	Active	2
1.1	The three dependencies	2
1.2	The measure	4
1.3	The task	7
1.4	What is measured on the human	9
2	Working	10
3	Peripheral	11
4	The clashes, listed	12
5	What crosses into the shared lexicon	12
6	Sources	12

1 Active

1.1 The three dependencies

role dependency

Definition: The study's name for a configured constraint between two teammates: who authorizes, who aggregates with whom, and who sees what. Three of the five CHART types are manipulated and two are left out.

Distinguishes it from: A role, which says what a teammate does. A role dependency says what a teammate cannot do alone. The triad has three roles in every condition, including the condition with no dependencies at all.

Operational test: Remove the teammate on the other end and ask whether the first teammate's action still resolves. If it does, that was a role and not a dependency.

In my project: The three manipulated are leader, pool and informational. Synchrony and temporal are not in this design, because the task has one judgement per episode and nothing to order or to time.

leader dependency

Definition: The requirement that one designated teammate's decision is the team's decision. In this study the human holds it, and the AI teammate's judgement is available as input rather than as a vote.

Distinguishes it from: The pool dependency, which is the alternative in the same manipulation. Under leader the human's answer is the answer. Under pool the human's answer is one of two inputs to a rule that produces the answer. The difference is not how much the human is told. It is whether the human can be outvoted by the aggregation.

Operational test: Have the human and the AI disagree and read the team's recorded judgement. If it is always the human's, the condition is leader.

In my project: The pre-registration defines it in the platform's own words, that an agent's execution depends on authorization from another agent, and then instantiates it as the human making the final scenario judgement. The predicted direction is conditional rather than flat: the leader dependency helps if the human is the more accurate of the two, and pooling helps if the LLM is.

Leader on the study, control on the platform

The pre-registration lists three role dependencies CHART offers and names the first one Leader, with the definition an agent's execution depends on authorization from another agent. That is word for word the platform's definition of Control.

The book chapter and `chart.md` both name the five types Control, Pool, Synchrony, Temporal, Informational. Control is the one with the approver, the requester, the approval-required action set, the escalation chain and the timeout policy. Leader appears in neither.

The implementation design of 3.23.26 names the same condition Control, with exactly the gloss the later document gives Leader: the human makes the final decision. The rename happened between March and the pre-registration.

The provenance of the new name is the decision-centralization literature the design is drawn from, where the contrast is between aggregated responses and a leader deciding. So the word came in with the manipulation, not with the platform, and the two are the same mechanism read from opposite ends. Control names the constraint on the one who must ask. Leader names the standing of the one who is asked. A results section that uses both interchangeably will be read as describing two conditions.

pool dependency

Definition: Here, the rule that combines the human's and the LLM's separate scenario judgements into one team judgement, so that neither teammate's answer is the team's answer on its own.

Distinguishes it from: The pool dependency as CHART defines it, where contributions accumulate toward a threshold and crossing it fires a composite action that none of the contributors executed. Here nothing is triggered. Two judgements arrive and a fixed rule reads them.

Operational test: Ask what happens when only one contributor acts. Under the platform's sense, nothing fires. Under this study's sense, the rule still produces an answer.

In my project: The aggregation is the part of the design with a formula behind it. Each agent supplies a signed confidence, the two are weighted by fitted reliability terms plus an intercept, and the sum goes through a logistic to give the probability that the scenario was an ongoing attack. The weights are fitted on a calibration set of human and AI judgements with known ground truth.

informational dependency

Definition: In this study, what the LLM teammate's summary contains, varied between an intuitive account of the pattern over time and a technical account naming the attack stages and the diagnostic signals.

Distinguishes it from: The platform's informational dependency, which is a filter on observations: a predicate that decides which of one agent's observations reach another, with a redaction level and a clearance condition. Here both conditions see the same IDS table. What differs is the description laid over it.

Operational test: Ask whether anything is hidden. If the same rows are on the screen in both conditions, the manipulation is resolution and not access.

In my project: The predicted direction is unusual and worth keeping in view. The expectation is that the human sees less than the two models do, and that seeing less raises complementarity by lowering the reading burden. That prediction is the opposite of the transparency assumption most of the literature runs on, and it is why the informational arm is in the design at all.

Informational names a filter and a resolution

The chapter and `chart.md`: informational dependencies govern access. A junior analyst gets filtered alert logs, a senior responder integrates across sources, and the parameters are redaction level, time gating and clearance.

The pre-registration: the informational manipulation is low resolution against high resolution summarization, both produced by the LLM over the identical table of events. The parameter is

the register of the summary.

Both are honestly informational, and the study's version is a difference in what is said rather than in what may be seen. Reporting it as restricting information, which the pre-registration does in one place, invites the reading that the human was denied data. The human was not.

role dependence, present and absent

Definition: The top-level manipulation, and the one the first research question turns on. Absent is the control condition, in which the three teammates complete every action independently with no interaction, and their decisions are aggregated for each task.

Distinguishes it from: The pool condition, which also aggregates. The difference is that in the control condition there is no summary, no discussion and no designated decider, so aggregation is the only thing joining them.

Operational test: Ask whether any teammate's screen changes because of another teammate. In the control condition, nothing does.

In my project: The design is described as partially nested, since role dependence subsumes the other two manipulations, which exist only when it is present. Two by two by two would give six cells. The nesting gives five.

1.2 The measure

complementarity

"The reason a pair can beat either member alone is that each holds knowledge it cannot fully externalize: the analyst's trained sense that something is off, the model's latent state."

website, FriendOrFoe, Why an aggregate score will not answer it

Definition: The property of a pairing whose joint output could not have been produced by either member working alone, because each contributes something the other does not hold.

Distinguishes it from: Joint performance, which is the number the team ends with. A pair can score well because one member carried it. Complementarity is a claim about where the answer came from, and the scoreboard does not record that.

Operational test: Ask what the other member would have answered on this item. If a counterfactual for each member alone is not recorded, the claim cannot be made.

In my project: The page states the obstacle in the same breath as the definition. A clean interface cannot be built across knowledge that cannot be handed over, so the interface between the two is the research problem rather than the implementation.

complementary team performance

"Complementary team performance obtains when the team's loss falls strictly below both the human's and the AI's alone"

website, FriendOrFoe, Why an aggregate score will not answer it

Definition: The field’s working definition, taken as stated: strict improvement over both solo baselines, on the same items.

Distinguishes it from: Complementarity as defined above. This one is a threshold on an aggregate and is therefore checkable. It is also, as the page says, a benchmark rarely reached, because the region where exactly one member is correct tends to be small.

Operational test: Compute three numbers on the same items and compare two inequalities. If either fails, the threshold was not met, whatever the team average looks like.

In my project: The page names the consequence that the study is designed around. A team can fail to show complementarity because the structure was wrong, or because there was none available to find, and an aggregate score does not separate the two.

the complementarity metric

Definition: The ratio the analysis plan uses: the team’s accuracy divided by the accuracy of the better of its two members, computed within a team over its five episodes of scenario judgement.

Distinguishes it from: The strict definition above, which is a yes or no on two inequalities. The ratio is continuous, so it can rank conditions that all fail the strict test, which is what makes it usable when complementary team performance is rarely reached.

Operational test: Read what is in the denominator. If it is the better member rather than the average of the two, the metric is this one, and a value of 1 means the team did no better than its best member.

In my project: Defined for two agents, so it is computed over the human and the LLM and over scenario judgements only. The ML teammate does not make scenario judgements and is therefore outside the metric even though it is inside the team.

team alignment

Definition: The proportion of decisions on which teammates gave the same answer. Two versions are recorded: agreement of all three on event classifications, and agreement of the human and the LLM on scenario judgements.

Distinguishes it from: Team accuracy, which compares answers to ground truth rather than to each other. Three teammates can align perfectly and all be wrong, and the design treats the relation between alignment and complementarity as something to correlate rather than to assume.

Operational test: Compute it without the answer key. If you cannot, you are computing accuracy.

In my project: Alignment is the closest available reading of whether the structure did anything to the process rather than to the score, which is the distinction the cluster’s open question on measuring complementarity keeps returning to.

the seam

“What is handed across the seam changes the team, in a direction that is not obvious in advance.”

website, FriendOrFoe, What the seam carries changes the team

Definition: Her word for the boundary between the human and the AI teammate, considered as

a thing with contents. What crosses it is a design variable, not a byproduct of the interface.

Distinguishes it from: The interface, which is where the crossing happens. The seam is what is chosen to cross: a label, a confidence, an explanation, or only the evidence.

Operational test: List what the human is shown, item by item. That list is the seam, and every item on it is a condition that could have been dropped.

In my project: The two manipulations are both seam manipulations read the right way. Information resolution changes what the summary says. Leader against pool changes what the human's own answer is allowed to do once it has crossed back.

over-reliance

Definition: Acceptance of the AI's answer at a rate the AI's accuracy does not support, so that the pair's outcome tracks the AI rather than the pair.

Distinguishes it from: Agreement, which is what it looks like. The two come apart only on the items where the AI is wrong, so a measure that does not split the trials by AI correctness cannot see it.

Operational test: Split by whether the AI was right and compare acceptance across the split. Flat acceptance is the finding.

In my project: The page states it as the warning the experiment is built to test: displaying the AI's explanation, confidence and label induced over-reliance, where showing the evidence alone left trust better calibrated. The source is the CHI 2021 study that also supplies the working definition above, where explanations raised the chance that people accepted the recommendation regardless of its correctness.

signed confidence

Definition: A single number carrying both which answer an agent gave and how sure it was, made by taking the confidence rating and giving it the sign of the binary choice.

Distinguishes it from: A confidence rating on its own, which is direction-free and cannot be summed across agents who disagree. Signing is what makes two agents' certainties combinable in one linear term.

Operational test: Ask what the value is when the agent is certain of the opposite answer. Under signed confidence it is the same magnitude with the other sign.

In my project: The human's is the self-report on a 0 to 100 scale, signed by their attack or not-attack judgement. The LLM's is derived rather than asked for.

reliability weight

Definition: The per-agent coefficient that says how much that agent's signed confidence should count, fitted rather than assumed.

Distinguishes it from: The confidence itself, which varies item by item. The weight is a property of the agent estimated across items, and it is what converts a stated certainty into an amount of influence.

Operational test: Ask what data the number came from. A weight needs a calibration set with ground truth. A confidence needs only the agent.

In my project: Two weights and an intercept, fitted by logistic regression on a calibration set of human and AI judgements. This is the concrete form of the combination rule the page says the literature fixes and measures.

perplexity as confidence

Definition: The LLM's certainty, read off the model's own surprise at the two candidate conclusions rather than from anything it says about itself. The difference between the surprise at attack and the surprise at no attack becomes the magnitude of its signed confidence.

Distinguishes it from: An LLM stating a number when asked how confident it is, which is a generated token and not a property of the distribution. The design takes the second route because the first is not a measurement.

Operational test: Ask whether the number would change if the model were asked to report it in a different format. A distributional quantity would not.

In my project: The design records this as a recommendation rather than as a fixed decision, and the reporting instructions still tell the LLM to rate its confidence from 0 to 100, so both routes are written into the same document.

1.3 The task

the triad

Definition: The three-member team: a low-level AI, a high-level AI and a human. Each holds a different part of the task and none of them does all of it.

Distinguishes it from: A dyad with a tool. All three are teammates in the design's own terms, and the division is by kind of judgement rather than by seniority. The ML model classifies events. The LLM summarizes and judges. The human judges.

Operational test: Count the agents whose output enters a recorded team decision. Three enter the event classification record. Two enter the scenario judgement.

In my project: The argument for the shape is a division of failure modes. The ML model classifies without hallucinating, the LLM matches patterns across sparse and disparate evidence, and the human supplies the reasoning that ties a sequence of events together. Placing the LLM inside a human-in-the-loop structure is what makes its opacity tolerable.

low-level AI and high-level AI

Definition: The two machine teammates, named by the grain of what they handle. The low-level one works on single events. The high-level one works on the sequence.

Distinguishes it from: A pipeline of two models, where the second consumes the first and the human sees only the end. Here the human sees the table, the low-level classifications and the high-level summary at once, and can disagree with any of them.

Operational test: Ask what the unit of output is. One row, or the whole episode.

In my project: The low-level teammate is a traditional machine learning classifier, chosen for the property that it does not invent. The high-level teammate is an LLM, chosen for the property that it can summarize across events that share no obvious feature.

event classification

Definition: The per-row decision, malicious or benign, over the sequential network events in one scenario's IDS table.

Distinguishes it from: The scenario judgement, which is one decision over the whole table. The two can disagree in both directions, and the design keeps them as separate dependent variables throughout.

Operational test: Count the decisions per episode. Twenty is classification. One is judgement.

In my project: Assigned to the low-level AI, with the other two teammates reviewing rather than re-deciding. In the control condition the human does it directly.

scenario judgement

Definition: The single binary call at the end of an episode: whether the sequence of events was an ongoing attack.

Distinguishes it from: Detecting an attack, which the event classifications already do row by row. The judgement is about whether the rows compose into one campaign, which is the thing the design claims the human is for.

Operational test: Ask whether the answer could be assembled from the rows without looking at their order. If it could, the task did not need the sequence.

In my project: Made by the human and the LLM separately, then resolved by whichever of leader or pool the condition specifies. It is the only decision the complementarity metric is computed on.

scenario, episode

Definition: One IDS table plus the decisions taken over it. The scenario is the content, the episode is the run.

Distinguishes it from: The zero-day scenario set, which is the library the tables are built from. Ten are documented and six were built.

Operational test: Ask whether the word refers to something a participant lives through or something a file contains.

In my project: Five per participant, randomly ordered, after one practice round.

How many events are in an episode

The pre-registration and the implementation design: twenty sequential events per scenario, twenty event classifications, one judgement, one confidence rating.

The built tables: thirty rows each, at a fifteen percent false alarm setting, seed 42, six scenarios assembled. The per-table counts are eleven malicious for WannaCry, nine for Data Theft and ShellShock, seven for Netcat Backdoor and passwd-gzip-scp, and none for the No Attack table, with four false alarms in every one of them.

Twenty is the number in the design documents and thirty is the number on disk, so one of the two moved and the other did not follow. This matters past bookkeeping, because the false alarm count is fixed at four per table rather than as a proportion, so a cut from thirty rows to twenty changes the false alarm rate the participant actually meets, which is the manipulation

the whole alert-triage literature this design borrows from turns on.

false alarm

Definition: An IDS event flagged by the detector that is benign in the ground truth. The tables carry it as a third label alongside malicious and benign, not as an error in the benign rows.

Distinguishes it from: A misclassification by a teammate. The false alarm is in the material before anyone sees it, and it is the reason the task is hard rather than a measure of how badly it was done.

Operational test: Look at the label column. Three values means false alarms are being modelled explicitly.

In my project: Fifteen percent by parameter, four rows per table in practice. The motivating fact from the alert-triage literature is that operational detectors run false alarm rates high enough to dominate the stream, which is what produces the fatigue the study is measuring load against.

zero-day

Definition: In this study, the condition the team is tested against: attacks outside the distribution the traditional model was fitted on.

Distinguishes it from: The four other senses the word carries across my files. This one is the experimental condition sense, and the shot count that belongs to it is a property of the label space rather than of the attack.

Operational test: Ask whose label space the novelty is relative to. The answer here is the low-level classifier's.

In my project: The full treatment is in `dict_zero_day.tex`, which enumerates all five senses and traces this one to the generation pipeline: public labeled traffic filtered by attack category, destination port and label, assembled into per-scenario tables.

1.4 What is measured on the human

cognitive load

Definition: The effort the task demands of the human participant, measured by self-report on a standard workload instrument at the end of the session.

Distinguishes it from: Time on task, which is behavioural and which the platform records anyway. The design carries both, and the fourth research question is whether structure lowers the reported one.

Operational test: Ask whether the number could have been computed from the logs. If it could, it is not the workload instrument.

In my project: Predicted to be highest in the condition with no dependencies, on the reasoning that a teammate doing the event classifications removes twenty decisions from the human. The informational arm is predicted to move it too, since the low resolution summary is less to read.

expertise

Definition: The participant’s prior cybersecurity knowledge, measured after the task and used as a covariate rather than as a screening criterion.

Distinguishes it from: Performance in the task, which it is expected to correlate with. Measuring it afterwards keeps it from priming the participant, and treating it as a covariate lets it modulate the effect of a dependency instead of only shifting the mean.

Operational test: Ask where the number enters the model. As a covariate it can carry an interaction. As a screen it cannot.

In my project: The expectation is an interaction rather than a main effect: the more expert the participant, the more they benefit from holding the final decision and from the technical summary.

AI acceptance

Definition: What the participant reports about working with the machine teammates, collected as open feedback and tailored to which teammates their condition actually had.

Distinguishes it from: Reliance, which is behavioural and read off the decisions. Acceptance is what they say afterwards, and the design keeps the two apart because the pairing of the two is the finding worth having.

Operational test: Ask whether the measure exists for a participant who never answered a question about the AI. Reliance does. Acceptance does not.

In my project: The questionnaire differs by whether the condition included the LLM, which is what stops the absent-teammate condition from generating opinions about a teammate nobody had.

2 Working

decision centralization

The contrast the design is built on, borrowed from a decision-making study of individuals against aggregated responses against a leader deciding. It is where the word leader comes from, and the two learning-side measures that came with it, how often a participant revises beliefs after feedback and how often they try something other than the best known option, are in the dependent variable suite as update rate and exploration rate.

CryWolf and the Impossible Travel dataset

The alert-triage platform and the alert set the task was first designed against, before the design moved to generated IDS tables. The dataset is authentication events describing logins from locations too far apart for the time between them, with true alarms and false alarms that both have ordinary explanations. The prior experiment on that platform varied false alarm rate between fifty and eighty-six percent and measured sensitivity, precision, time on task, workload and decision confidence, which is where most of this design’s dependent variables come from.

the earlier binary classification design

The version before the triad, with three conditions taken from the same decision centralization contrast. Decentralized, where every member classifies and the average is taken. Centralized, where a leader decides from the other two members' answers. Individual, with no dependencies. It is worth keeping because it names centralized as pool and control together, which is the reading under which leader and control stop competing.

partially nested design

The structure of the analysis: role dependence as the outer factor, with authority and information resolution existing only inside its present level. Two by two by two is six cells and this design has five, and the analysis carries the nesting explicitly rather than treating the control condition as a fourth cell of a full factorial.

the vulnerability line

The earlier framing of the same project, before it narrowed. The question then was which features of a configured team an adversary could exploit: flooding a control edge to stall it, an insider withholding a pool contribution, deception to break a synchrony window, delay against a temporal prerequisite, and false or selective disclosure against an informational edge. Nothing of it is in the current design, and it is the natural second study.

3 Peripheral

alert triage and escalation

The operational task the classification stands in for: deciding which alerts are worth a person's attention and which get passed upward. The design's binary malicious or benign call is the compressed form of it.

IDS

The detector producing the events. It stores attributes of network activity and checks them against known signatures, rule violations and statistical anomalies, which is what fixes its blind spot as the thing a zero-day condition is defined against.

multi-stage attack

The property the scenario judgement is asking about, where individually unremarkable events compose into one campaign. It is the reason the task has a second decision layered over the first rather than only rows to label.

4 The clashes, listed

The disagreement	The entry it sits in
Leader in the study, control on the platform	leader dependency
Informational as a filter, or as a resolution	informational dependency
Pool as an aggregation rule, or as a quorum trigger	pool dependency
Twenty events in the design, thirty in the tables	scenario, episode

The first three are one problem seen three times. This study borrowed three dependency names from CHART and gave two of them different mechanisms and the third a different name, and a reader who learned the vocabulary from the chapter will misread all three. The fix is not to renormalize the study to the platform, because the study’s mechanisms are the ones the data will be collected under. The fix is to say, once, in the methods, which sense is in force.

5 What crosses into the shared lexicon

Terms this project uses that the whole Human-AI Complementarity cluster uses. Their cluster sense goes in the shared file, and the entries above carry only what FriendOrFoe fixes about them: complementarity, interdependence, authority, trust calibration, over-reliance, the five dependency types, and the tacit knowledge argument that opens both this page and the cluster README.

Two of them are fixed here and nowhere else in the cluster. The complementarity metric is the only operational definition of complementarity anywhere in my files, and it is a ratio over two agents on one decision type, which is narrower than the word is used elsewhere. The seam is her name for the human-AI boundary considered as contents rather than as a surface, and it is the term that makes information resolution and the leader-pool contrast the same manipulation seen twice.

One term comes in from outside the cluster and should stay attached to its source. Zero-day is defined in `dict_zero_day.tex` across five senses, and this page uses the experimental-condition sense only.

6 Sources

- Every quoted sentence is from the FriendOrFoe page of the website. The locator on each entry says which section.
- The three manipulated dependencies, their definitions, the predicted directions, the triad’s role assignment, the dependent variables, the analysis plan and the complementarity ratio are from the pre-registration document in the FriendOrFoe folder.
- The aggregation formula, the signed confidence construction, the perplexity route to the LLM’s confidence and the two summary registers are from the implementation design of 3.23.26 in the same folder’s meeting notes.

- The earlier three-condition design, the decision centralization contrast, the alert-triage platform and dataset, the dependent variable suite and the vulnerability line are from the Role Interdependence Experiment slides and the Task Design document.
- The event counts, the false alarm setting and the per-table label distributions are read from the assembly summary and the tables in `IDS_zero_day_generation`.
- The five dependency types, their parameters and the platform’s definitions of control and informational are from the CHART book chapter manuscript and from `chart.md`. They belong properly to `hac_chart.tex` and appear here only where this study renames or reuses them.
- The working definition of complementary team performance and the finding about explanations and acceptance are from Bansal et al., CHI 2021, read from the paper’s own statement of both.
- The three tiers, the three fluency levels and the four fields are from the Tech Fluency page. The reason a term whose two uses disagree belongs inside its entry is in `lexicon_method.tex`, in this folder.